



Patrones de comportamiento

Patrón Interpreter

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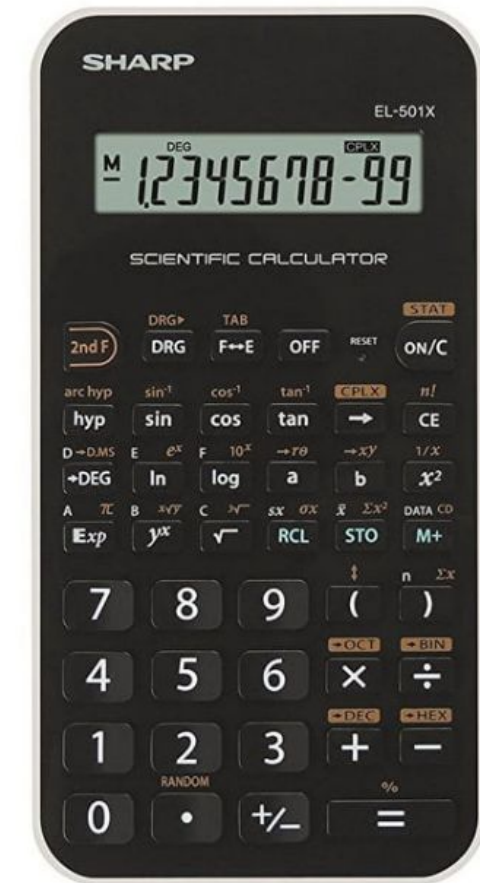
¿Qué hace?

Evalúa gramáticas
personalizadas y
ejecuta lenguajes de
dominio específico.

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Es como una calculadora

- Análisis: Interpreta Reglas estrictas
- Jerarquía: Evalúa precedencia matemática
- Ejecución: Calcula resultado final



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Árbol sintáctico

Es la estructura interna básica conformada por:

- Nodos terminales básicos
- Operadores como ramas



Abstract syntax tree

An **abstract syntax tree** (AST) is a data structure used in computer science to represent the structure of a program or code snippet. It is a tree representation of the abstract syntactic structure of text (often source code) written in a formal language. Each node of the tree denotes a construct occurring in the text. It is sometimes called just a **syntax tree**.

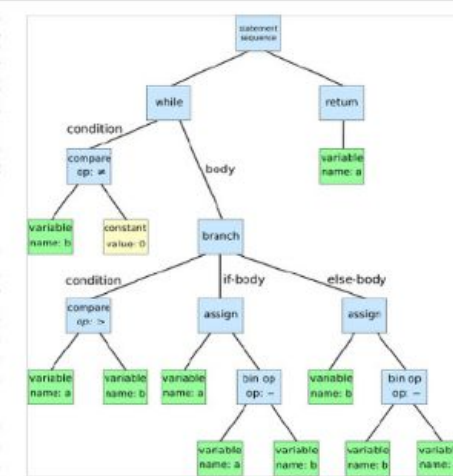
The syntax is "abstract" in the sense that it does not represent every detail appearing in the real syntax, but rather just the structural or content-related details. For instance, grouping parentheses are implicit in the tree structure, so these do not have to be represented as separate nodes. Likewise, a syntactic construct like an if-condition-then statement may be denoted by means of a single node with three branches.

This distinguishes abstract syntax trees from concrete syntax trees, traditionally designated **parse trees**. Parse trees are typically built by a parser during the source code translation and **compiling** process. Once built, additional information is added to the AST by means of subsequent processing, e.g., **contextual analysis**.

Abstract syntax trees are also used in **program analysis** and **program transformation** systems.

Application in compilers

Abstract syntax trees are data structures widely used in compilers to represent the structure of program code. An AST is usually the result of the syntax analysis phase of a compiler. It often serves as an intermediate representation of the program through several stages that the compiler requires, and has a



An abstract syntax tree for the following code for the Euclidean algorithm:

```
while b ≠ 0:
  if a > b:
    a := a - b
  else:
    b := b - a
return a
```

Componentes del patrón



Contexto

Estado global
compartido.



Expresión Base

Interfaz principal
evaluadora.



Nodo Terminal

Valores absolutos
finales.



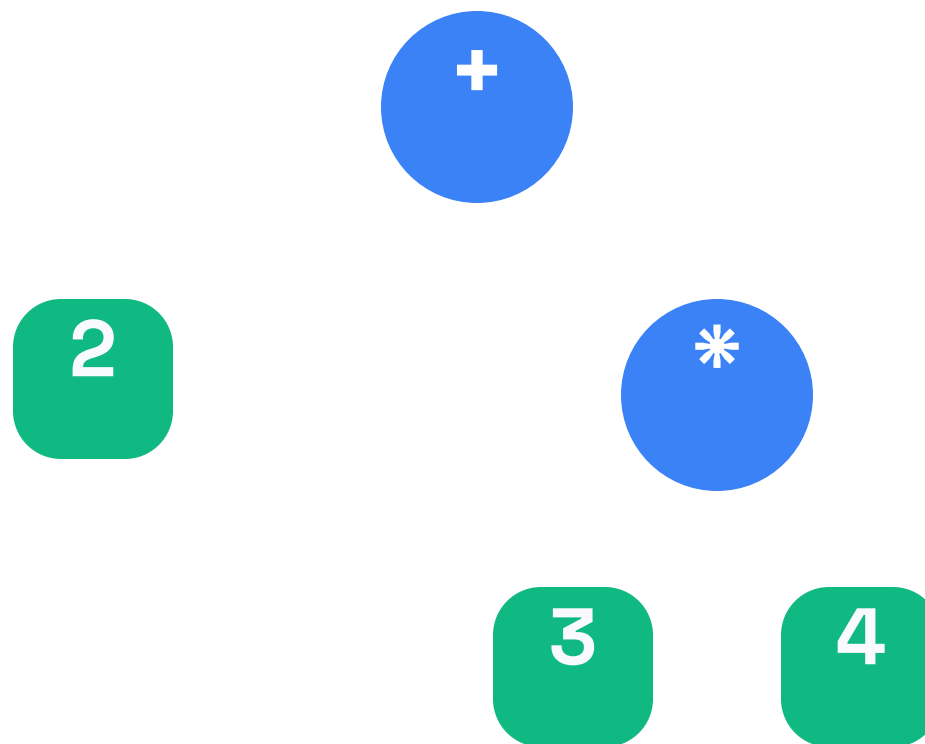
Nodo Compuesto

Operaciones
lógicas
intermedias.

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Mapeo de la expresión

$$2 + (3 * 4)$$



Diseño Modular

1. Cada regla gramatical propia
2. Polimorfismo en evaluación
3. Recursión natural integrada

```
class Suma : public Expr {  
    Expr* izq;  
    Expr* der;  
public:  
    int interpretar() {  
        return izq→interpretar() +  
            der→interpretar();  
    }  
};
```

Aplicaciones prácticas

- Motores SQL
- Motor de reglas



iStock
Credit: Kanls

A la V



+ Ventajas

- Extensión gramatical sencilla.
- Reglas modulares aisladas.
- Mapeo directo sintáctico.

- Desventajas

- Explosión masiva de clases.
- Desempeño temporal deficiente.
- Inviabile en gramáticas complejas.

Image Sources



https://i5.walmartimages.com/seo/Sharp-EL-501XBWH-Scientific-Calculator-Black_7e0545c1-e060-42e4-8acd-11f2f8be5a96.468a16323f4c8318376f536daf595c50.jpeg?odnHeight=768&odnWidth=768&odnBg=FFFFFF

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¡Gracias!